

Surface area of revolution

Key Points

The curved surface area of revolution for $y(x)$ is

$$\int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

The curved surface area of revolution for $x(t), y(t)$ is

$$\int_{t_1}^{t_2} 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

With both revolved around x -axis. Note you may need to add the top and bottom for the surface area of the shape

Basic Skills

Q1. State the volume of revolution for $y(x)$ around the x -axis

Q2. For the function $y = 2x$

a) Sketch the graph and determine what 3D shape is formed by a revolution around the x -axis

b) Show that the integral for the volume of the shape formed between $x = 0, 3$ is $\int_0^3 4\pi x^3 dx$ and hence state the volume

c) Show that the integral for the surface area of surface formed between $x = 0, 3$ is $\int_0^3 4\sqrt{5}\pi x dx$ and hence find the surface area of the curved

d) Confirm your volume and surface area by using geometric formulae for the cone $V = \frac{\pi r^2 h}{3}, S = \pi r(r + l)$

Competency Skills

Q3. Explain if it is possible to form a square based pyramid by revolving a function around the x -axis

Q4. Find the curved surface area of the paraboloid, formed by revolving the general parabola with equation $y^2 = 4ax, x \in [0, a]$ for any $a \in \mathbb{R}$

Q5. The asteroid is given by $x(\theta) = a \cos^3 \theta, y(\theta) = a \sin^3 \theta$ for $0 \leq \theta \leq 2\pi$ for some a

a) Sketch the asteroid and state the symmetries

b) Calculate the surface area for the revolved asteroid in terms of a

Q6. A frustum of a paraboloid is formed by the parametric equation $x = at^2, y = 2at$ between $x = a$ and $x = 4a$, work out the total surface area of the frustum

Q7. A period of the cycloid $x = a(t - \sin t), y = a(1 - \cos t)$ is revolved work out the surface area

Advanced Skills

Q8. A student states that "The volume and surface area of any infinitely long shape must be infinity as it is similar to multiplying by infinity"

a) Using $y = \frac{1}{x}$ to form a trumpet shape for $x \in [1, \infty)$ show that the volume is finite

b) Using $y = \frac{1}{x}$ to form a trumpet shape for $x \in [1, \infty)$ show that the curved surface area can be given by the integral $2\pi \lim_{k \rightarrow \infty} \int_1^k \frac{\sqrt{1+x^{-4}}}{x} dx$

c) Hence by using an appropriate Maclaurin's expansion deduce if the surface area is finite or not

d) What conclusion could we draw about the surface area of the trumpet formed by $y = \frac{1}{x^2}, x \in [1, \infty)$

Extension

Q9. Consider if you think the statement "All 3D shapes of finite area have finite volume" is true

Hint: Look at Gabriel's horn, formed by $\frac{1}{x^2}$